Vidusha Sanidu

AI/ML Researcher | Computer Science Graduate

Phone: +94 76 041 6256 **Email:** vidushsanidu07@gmail.com
LinkedIn: linkedin.com/in/vidushasanidu
GitHub: github.com/VidushaSanidu **Portfolio:** viduzx.xanvia.tech

PROFILE

Computer Science graduate and Research Assistant with research and engineering experience in federated learning, machine learning, distributed AI systems, and data-driven optimization. Experienced in designing, implementing, and evaluating ML systems using real-world datasets, with particular experience in decentralized model training, federated aggregation, heterogeneous client environments, and applied AI systems. Interested in advancing research in federated and privacy-preserving machine learning, distributed optimization, and resource-efficient intelligent systems.

RESEARCH INTERESTS

Federated Learning • Privacy-Preserving Machine Learning • Distributed and Decentralized Machine Learning • Distributed Optimization • Resource-Efficient AI • Multi-Agent System

EDUCATION

University of Peradeniya, Sri Lanka (2021 – 2025)
BSc (Hons) in Computer Science — Final GPA: 3.67

ACADEMIC RESEARCH EXPERIENCE

Research Assistant (2026 May - Current)
University Research Council (URC), University of Peradeniya
Research Title: Effective AI-Based Resource Distribution Management System in Sri Lankan Hospitals

- Contribute to a multidisciplinary research project developing AI-driven resource distribution and decision-support methods for Sri Lankan hospitals, with an emphasis on data-driven management under resource constraints.
- Evaluated an ensemble learning framework for predictive hemodialysis management using longitudinal real-world clinical data collected from the Renal Care Unit, Teaching Hospital Peradeniya.

Personalized Pedestrian Trajectory Prediction using Federated AI
Final Year Thesis – University of Peradeniya [Manuscript under peer review in the Ceylon Journal of Science (CJS)]

- Implemented a custom Light-SIAT (Spatial-Temporal) deep learning architecture integrated with a Federated Learning (FedAvg/FedAtt) framework to evaluate decentralization performance without centralized data pooling.
- Investigated the impact of decentralized training and client configurations on predictive performance, local updates, and the trade-off between model utility and data decentralization.

- Studied federated model aggregation and personalization approaches for improving performance across heterogeneous clients and distributed data environments.

Domain Adaptation & Local Vehicle Detection in Sri Lankan Transit
Research Project under Dr. Samal Dharmarathne

- Investigated domain shift vulnerabilities in globally pre-trained vision models when applied to region-specific vehicle classes (e.g., three-wheelers, localized buses).
- Gathered local dataset images captured across diverse lighting and shadow conditions; applied layer-freezing and transfer learning strategies on YOLOv8 to improve detection accuracy in low-light environments.

AI & ML ENGINEERING EXPERIENCE

- **Agentic Academic Assistant & Multi-Tool RAG Pipeline**
  - *Stack:* Python, FastAPI, LangChain, ChromaDB, Gemini Pro API, OCR Modules, WebSockets, React.js
  - Designed an interactive LLM-driven agent system designed to automate academic path generation based on student transcripts, state pathway guidelines, and institutional course lists.
  - Developed a dual-agent workflow, featuring a General Q&A Agent and a Study Plan Generator Agent using LangChain tools to dynamically route queries between vectorized documents and real-time computation modules.
  - Integrated OCR parsing pipelines for automated unstructured document extraction and implemented safety guardrails (rate limiting, context window optimization, hallucination mitigation).
- **Echo-IC: Voice Synthesis & Analysis Pipeline**
  - *Stack:* XTTS Model,OpenAI Whisper API, PyTorch, Python, FastAPI, Redis / Asynchronous Job Queues, Docker
  - Developed an deep learning pipeline with speech-to-text transcription, LLM-based voice analysis, and zero-shot voice cloning.
  - Built a CI/CD pipeline for the Docker-containerized system, automating container image building and deployment for a reproducible ML application delivery workflow.

INDUSTRY & PROFESSIONAL EXPERIENCE

AI & Automation Engineer - Intern (Sep 2025 – Jan 2026)
Taino Spain SL (Remote - Spain)

- Designed production-grade Retrieval-Augmented Generation (RAG) systems using LLM backends and vector databases to parse complex, unstructured technical databases.
- Handled production challenges including LLM hallucination filtering, rate-limit mitigation, and custom agent tool definition.

TECHNICAL SKILLS

- **Machine Learning:** PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, Random Forest, YOLO, Predictive Modeling, Transfer Learning, Model Fine-tuning

- **Distributed and Decentralized AI:** Federated Learning, FedAvg, Distributed Model Training, Client-based Learning, Model Aggregation, Distributed AI Systems
- **Programming Languages:** Python, Java, C/C++, TypeScript, JavaScript, SQL, LaTeX
- **Backend, Cloud & DevOps:** FastAPI, Express.js, NestJS, Node.js, Microservices, Docker, Redis, AWS (S3, Lambda), Linux

LEADERSHIP & ACHIEVEMENTS

- **Vice President (2024–2025):** Computer Society, University of Peradeniya (CSUP). Organized technical workshops and industry guest lectures.
- **Webmaster (2024–2025):** IEEE Computer Society Student Branch, University of Peradeniya.
- **Top 15 Finalist:** Mini Hackathon in Data Science, Statistics Circle, University of Colombo.
- **6th Place:** Manthra CTF Cybersecurity Competition, IEEE CS NSBM Chapter.
- **Winner:** KickStart Weekend Boot Camp (2022), CGU University of Peradeniya.

REFERENCES

Prof. Saluka Kodituwakku
Senior Professor
Department of Statistics and Computer Science
University of Peradeniya
Relationship: Research Supervisor

Dr. S. Maheswaran
Senior Lecturer
Department of Operations Management, FOM
University of Peradeniya
Relationship: Principal Investigator